Rochester Institute of Technology
B. Thomas Golisano College of Computing and Information Sciences
Master of Science in Human–Computer Interaction

Designing and Evaluating Usability of AI-Enhanced Context-Aware Adaptive User Interfaces for Mobile App Library/Drawer

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This Capstone Project has been reviewed and approved by the undersigned members of the supervisory committee. The signatures below certify that the project meets the requirements of the Master of Science in Human–Computer Interaction programme at the Rochester Institute of Technology.

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Abstract

Mobile operating systems currently rely on static, grid-based app galleries that force users to manually organize, search, and retrieve applications. As the number of installed applications grows, this static approach increases cognitive load and friction, particularly because user needs and contexts shift dramatically throughout the day. In this study, I investigate the usability of AI-driven adaptive user interfaces (AUIs) for mobile app galleries, focusing on addressing the limitations of static library/gallery interfaces through personalized, context-aware design. Through a mixed-methods approach — a participatory design workshop phase followed by high-fidelity usability testing of a functional web prototype (N=12) — I evaluated a dynamic interface that shifts between distinct contextual states: Neutral/Default, Morning Routine, Work/Focus Mode, and Evening/Relax Mode.

I measured task completion times, error rates, and qualitative perceptions of agency, privacy, and system transparency. Key findings indicate a significant learning effect, with task completion times decreasing by an average of 42% across repeated accesses within the same contextual state. While users expressed high satisfaction with the visual design and the contextual page model (mean SUS: 74.5), the study also highlights critical usability gaps: the necessity for explicit AI transparency indicators, the demand for robust manual override and error-recovery mechanisms (e.g., folder creation), and essential accessibility considerations for colour-vision deficiencies. My findings provide actionable design guidelines for integrating probabilistic AI behaviours into deterministic mobile operating systems without sacrificing user agency.

Keywords: Adaptive User Interfaces, Context-Aware Computing, Mobile App Gallery, Artificial Intelligence, Usability Testing, Human-Computer Interaction, Digital Wellbeing.

Contents

1 Introduction

Do you often struggle to locate apps on your phone during hectic moments? The app gallery — a foundational element of most smartphone operating systems, commonly presented as a grid or list of installed applications — exhibits limited adaptability to the user's evolving behaviour or contextual priorities [1]. Despite its prevalence, this static architecture creates navigational friction, as users are compelled to interact with an interface that lacks responsiveness to their individualised needs and fluctuating routines. While the app gallery remains ubiquitous, it offers limited personalisation despite the increasing sophistication of modern mobile ecosystems (for reference, Figure 1 illustrates the standard categorical grid that this study takes as its baseline).

Figure 1: Schematic illustration of a static, grid-based app gallery (left) and the navigational friction it creates (right). As application libraries grow beyond 50–100 apps, spatial memory deteriorates and users increasingly resort to search [2, 3, 4].

Imagine a personalised app gallery that adapts dynamically, organising apps based on user preferences, frequencies of usage, and contextual needs. Such systems are possible through Adaptive User Interfaces (AUIs), which leverage user behaviour analysis and predictions to improve accessibility and efficiency [1, 5]. AUIs tailor menus to specific needs, reducing cognitive loads and improving navigation efficiency. Current mobile operating systems have begun to explore similar concepts — iPhones offer Siri Suggestions, Android provides smart app libraries — but these implementations remain limited in adaptability and personalisation capabilities, presenting a significant opportunity for advancement (see Figure 2 for an example of the current single-row Siri Suggestions implementation this study seeks to go beyond).

Figure 2: Existing single-row adaptive implementation: iOS Siri Suggestions, showing a predictive row of apps and shortcuts at the top of a home screen page while the remainder of the interface stays in a static grid. Current OS implementations are constrained to this single-row approach and do not adapt the full-page layout to macro-level user contexts [6, 7, 8].

Yet tailoring the app gallery is not without challenges. Cockburn et al. [8] demonstrate that menu performance depends heavily on the trade-off between speed and predictability. Building on this, I ask: how can adaptation avoid introducing unpredictability or user confusion? How can menus balance personalisation with simplicity and speed? Studies have highlighted risks where excessive adaptiveness — such as frequent reorganisations — could diminish usability [5]. This capstone project introduces an innovative concept: an adaptive app gallery that intelligently reorganises itself based on user behaviour, preferences, and contextual factors. By adopting iterative usability testing, participatory design workshops, and data-driven insights, the project seeks to reimagine mobile navigation beyond conventional static designs.

1.1 Research Questions

This study is guided by the following primary research questions, grounded in the IRB-approved protocol and refined through the participatory design phase:

RQ1 (Efficiency and Learnability): What key design principles and context-aware strategies can be employed to optimise the adaptive app gallery for seamless personalisation while maintaining predictability and minimising cognitive load for users?

RQ2 (Trust and Transparency): What impact does the ability of AI to organise applications based on usage have on user satisfaction and efficiency? What specific interface elements are required to establish user trust and communicate the rationale behind AI-driven spatial displacements?

RQ3 (Agency and Control): Which manual override and error-recovery mechanisms are most critical to mitigating the cognitive friction introduced by probabilistic UI adaptations?

2 Related Work

The design of an adaptive mobile app gallery draws on several intersecting areas of research: mobile app usage and organisation strategies, context-aware computing, the tension between predictability and adaptability, explainable AI in consumer interfaces, digital wellbeing, and accessibility in adaptive design.

2.1 Mobile Application Usage and Organisation Strategies

Research on mobile app usage consistently reveals a tension between growing app libraries and static organisational interfaces. Early large-scale work established that usage is highly fragmented and context-dependent, occurring in short bursts throughout the day [2]. Users respond to app clutter through three main organisation strategies — frequency-based, category-based, and spatial memory-based arrangements [3] — yet all three impose a continuous maintenance burden. As libraries exceed 50–100 apps, spatial memory deteriorates and users shift from visual scanning to query-based retrieval [4]. Taken together, this body of work establishes that static app drawers are increasingly misaligned with real-world usage patterns, motivating the adaptive approach I investigate here.

2.2 Context-Aware Computing and Predictive Interfaces

Context-aware computing uses sensor streams — time, location, accelerometer, network state — to infer user intent and surface relevant content proactively [9]. Prior implementations have demonstrated that temporal usage patterns can predict the next app a user will open, and that predictive ranking measurably reduces the physical actions required to launch apps [6, 7]. However, these gains are consistently constrained to a single predictive row, leaving the broader interface unaddressed. More critically, when systems attempt aggressive icon displacements within a single grid, user satisfaction declines sharply due to disrupted spatial consistency [8]. These findings directly motivated my decision to adapt at the macro-state level rather than within a single grid — the core architectural principle of the Contextual Page Model.

2.3 Balancing Predictability and Adaptability

The fundamental tension in adaptive UI design — between system automation and user control — is well-documented. Gajos et al. [5] and Findlater and Gajos [1] both found that adaptation benefits are frequently outweighed by the performance cost of disrupted spatial memory, particularly when changes occur within a single grid. The threshold for predictive accuracy at which adaptation yields net benefit is rarely achieved in real-world conditions [5]. Horvitz's mixed-initiative principles [10] argue that any intelligent system must preserve the user's ability to directly invoke and terminate automation; when AI silently repositions content without consent, it violates system-status visibility norms and triggers automation surprise [11]. My study builds on these findings by testing whether macro-level contextual page transitions can resolve this tension in a way that micro-level icon displacement cannot.

2.4 Explainable AI in Consumer Interfaces

A convergent body of work on transparent personalisation demonstrates that users require intelligible feedback about interface changes to build calibrated trust [12, 13, 14]. Amershi et al. [15] formalised this into 18 guidelines for Human-AI Interaction, with the core mandate that systems make clear why they did what they did. Complementary empirical work shows that explanations improve both system intelligibility and user acceptance [13], and that even low-cost explanations in deployed systems produce meaningful trust gains [14]. Despite these established guidelines, current mobile OS implementations treat the adaptive app drawer as a black box. My study directly evaluates user responses to both transparent and opaque adaptive behaviours, examining how concepts like visual muting and contextual page labels affect perceived interface ownership.

2.5 Digital Wellbeing and Attention Management

The integration of digital wellbeing features into adaptive interfaces addresses a growing concern about smartphone overuse and attention fragmentation. Hiniker et al. [16] demonstrated that deliberate friction mechanisms — such as restricting access to high-dopamine applications during productive hours — can effectively reduce impulsive app engagement without triggering anxiety. This contrasts with hard-blocking approaches, which participants in prior studies have described as causing disorientation and distress. The present study integrates wellbeing mechanics through two distinct friction strategies: the Context Lock, which places social media applications in a visually restricted state during morning hours, and Visual Muting, which reduces the attentional salience of entertainment applications during work periods.

2.6 Accessibility in Adaptive UI Design

Adaptive interfaces introduce novel accessibility challenges when their visual cues rely solely on colour or saturation changes to communicate system state. Flatla and Gutwin [17] documented that colour-vision deficiencies significantly impair users' ability to interpret interfaces that depend on hue or saturation distinctions. The Web Content Accessibility Guidelines (WCAG) 2.1, specifically Success Criterion 1.4.1 (Use of Colour), explicitly require that colour must not be used as the only visual means of conveying information, indicating an action, prompting a response, or distinguishing a visual element [18]. This principle is particularly relevant to adaptive app galleries that employ greyscale rendering as a muting signal, an approach that, while effective for users with standard colour vision, creates severe navigation barriers for individuals with conditions such as deuteranopia.

2.7 Commercial Context-Aware OS Features and Their Limitations

Contemporary mobile operating systems have introduced several context-sensitive features that partially address the limitations of static app galleries. Situating the Contextual Page Model against these commercial implementations is necessary to establish the precise nature of the research gap this study fills. *iOS Siri Suggestions* (introduced with iOS 10) predicts the four most contextually relevant apps based on usage history, time, and location, surfacing them in a single predictive row at the top of the App Library [6, 7]. While this demonstrates that OS-level context-awareness is technically feasible, it is constrained to a single row: the remainder of the interface — the categorical grid, the page structure, the visual treatment of individual apps — remains entirely static. *iOS Focus Mode* (iOS 15, 2021) allows users to create named contextual profiles — Work, Personal, Sleep, or custom — that suppress notifications from non-permitted sources and can show or hide pre-configured home screen pages [19]. Unlike the Contextual Page Model, Focus Mode requires extensive manual configuration; it executes user-authored rules rather than inferring context automatically. More critically, Focus Mode does not alter the internal layout or content of any visible page: apps remain in the same positions with the same visual treatment, no contextual widgets are injected, and no novel layout paradigms are introduced. When a Focus activates, no rationale is communicated to the user — the system is opaque by design.

iOS App Library (iOS 14, 2020) automatically organises all installed applications into fixed category folders — Social, Entertainment, Utilities, Recently Added — on a dedicated end page [20]. Although the Suggestions row within App Library surfaces four apps based on usage prediction, the category structure itself is static and does not adapt to time of day, inferred context, or user state. An Entertainment folder presents identical content at 7 AM and at 10 PM.

Screen Time (iOS 12, 2018) and its Android equivalents provide usage reporting and allow users to impose daily time limits on app categories or schedule system-wide Downtime windows. These features are purely restrictive: they reduce access to applications but do not reorganise the interface to surface relevant alternatives, inject contextual affordances, or adapt layout to user intent. Furthermore, hard-blocking approaches — removing or locking apps entirely — have been shown to cause user disorientation and anxiety [16, 21], motivating the softer friction mechanics employed in the present study.

Taken together, these commercial implementations share four fundamental limitations that the CPM is designed to address. First, they operate at the notification management and page-visibility layer, leaving the internal page architecture — layout structure, widget composition, visual treatment of individual apps — entirely unaddressed. Second, those that require configuration (Focus Mode, Screen Time) impose a high setup burden that limits adoption among general users. Third, none introduce novel layout paradigms responsive to contextual intent: no commercial system replaces a categorical grid with a horizontally browsable intent-based layout for evening relaxation, or injects a meeting-aware Productivity Slab widget during work hours. Fourth, none provide explicit

Table 1: Systematic comparison of commercial iOS/Android context-aware features against the Contextual Page Model (CPM) across ten functional dimensions. ✓ denotes a CPM capability absent from all commercial implementations evaluated.

Dimension	Siri Sugg.	iOS Focus Mode	App Library	Screen Time	CPM (This Study)
Automatic context inference	Partial	No	No	No	✓ Full (time, calendar, alarm, GPS)
Full-page layout restructuring	No	No	No	No	✓
Novel layout paradigms	No	No	No	No	✓ (horizontal intent, Slab)
Contextual widget injection	No	No	No	No	✓ (Morning Widget, Prod. Slab)
Visual state of apps	No	No	No	No	✓ Greyscale muting
Wellbeing friction	No	No	No	Blocking	✓ Context Lock
AI transparency / rationale	No	No	No	Reports only	✓ Plain-language
Zero user configuration	Yes	No	Yes	No	✓
Spatial memory preserved	Breaks	N/A	N/A	N/A	✓ (position preserved)

AI transparency: when these systems change the interface, the change is unexplained. Table 1 summarises these distinctions across ten functional dimensions.

3 System Design: The Contextual Page Model (CPM)

To evaluate user interactions with dynamic mobile environments, I developed a high-fidelity, functional web-based prototype using React and Vite, styled to emulate a native mobile operating system interface consistent with iOS 17 aesthetics. Because a fully functional, context-aware OS-level launcher requires deep system integrations unavailable for rapid prototyping, I deployed the application using a Wizard of Oz methodology. A hidden, moderator-controlled Simulation Menu allowed me to perform real-time manipulation of the interface's contextual states during usability testing.

The Contextual Page Model abandons the traditional single-grid app drawer in favour of four distinct, dynamically shifting macro-states. This architectural decision — to adapt at the macro level rather than displacing individual icons within a single grid — was directly motivated by evidence from Findlater and Gajos [1] and Gajos et al. [5] that micro-level adaptations break spatial memory and frequently reduce efficiency.

State 0 (Neutral/Default) is a baseline static grid of familiar applications organised categorically, accompanied by a persistent universal search bar. It serves as both the onboarding state and the always-reachable fallback — a deterministic escape hatch from any contextual state (see Figures 4a and 6a).

State 1 (Morning Routine) is triggered hypothetically by waking hours and alarm dismissal. This state promotes productivity and communication tools to the top row (see Figure 4b). Crucially, it introduces a Context Lock feature — placing high-dopamine social media applications into a visually restricted folder to discourage morning doom-scrolling, a deliberate friction mechanic for digital wellbeing [21]. This feature was independently proposed by multiple design workshop participants before it was implemented in the prototype.

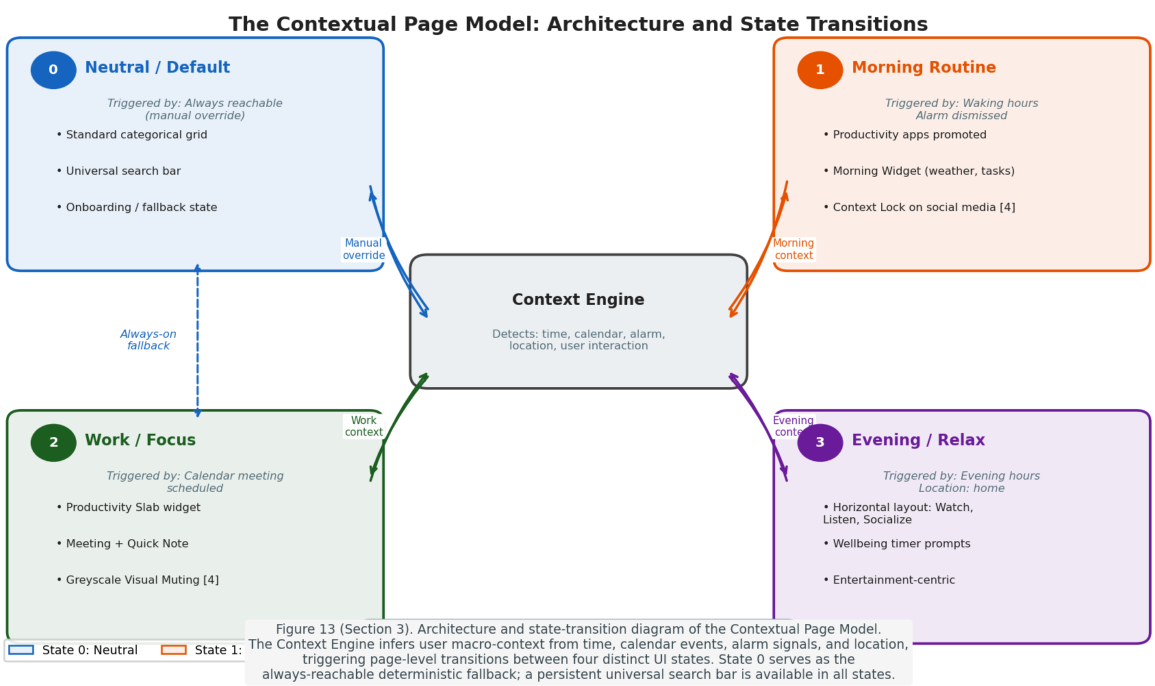
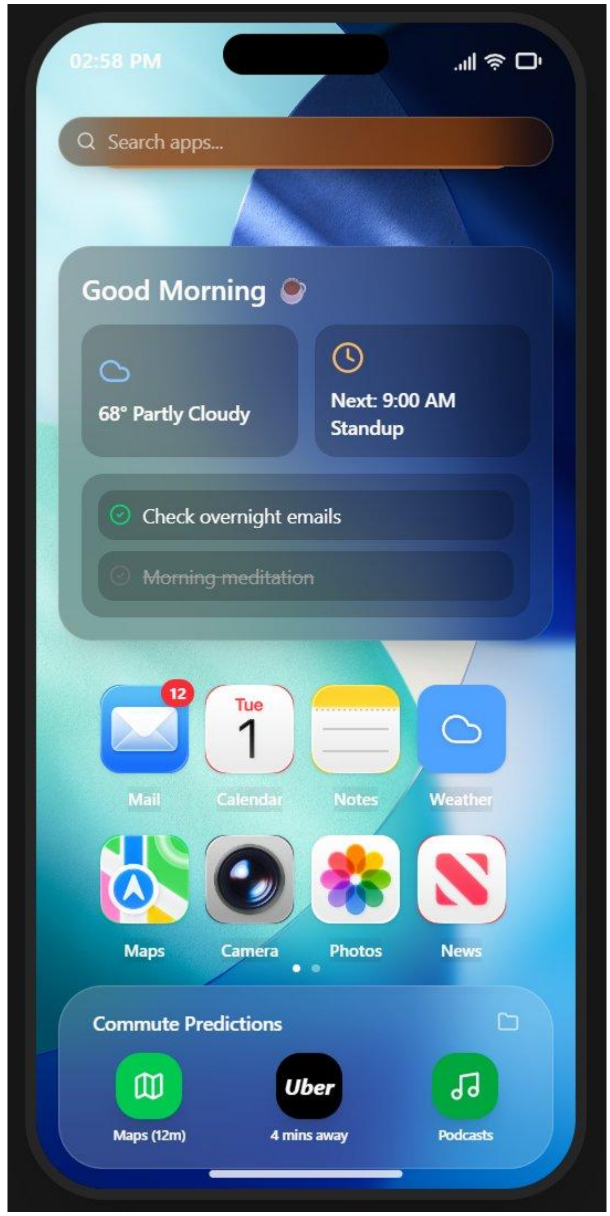


Figure 3: Architecture and state-transition diagram of the Contextual Page Model. The Context Engine infers macro-context from time, calendar, alarm, and location signals, triggering transitions between four distinct UI states. State 0 (Neutral) is always reachable as a deterministic fallback; a universal search bar persists across all states (see Figures 4a–7b for prototype screenshots).

State 2 (Work/Focus Mode) surfaces a Productivity Slab widget displaying upcoming meetings and a notes scratchpad. To maintain spatial memory while reducing distraction, non-work applications are not hidden but are visually muted — rendered in greyscale — to make them less visually appealing while preserving their physical location. The greyscale approach was preferred by workshop participants over total app concealment, which they described as causing panic (see Figure 5a).

State 3 (Evening/Relax Mode) radically shifts the structural layout from a strict grid to a media-centric, horizontally scrollable format categorised by intent: Watch, Listen, and Socialise. It also integrates periodic digital wellbeing prompts for extended social media engagement (see Figures 5b and 7b). This state was directly requested by design workshop participant P3, who sketched themed category swipes before seeing any prototype.

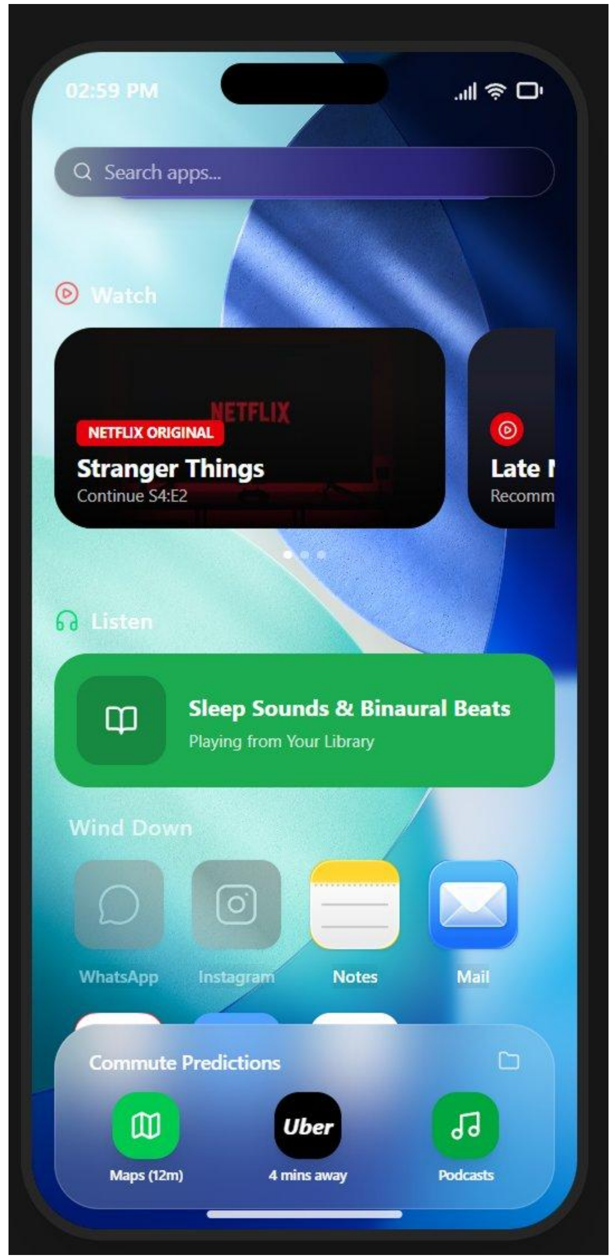
(a) State 0: Neutral/Default — standard categorical grid with universal search bar.



(b) State 1: Morning Routine — weather/task widget, promoted productivity apps, Context Lock on social media.

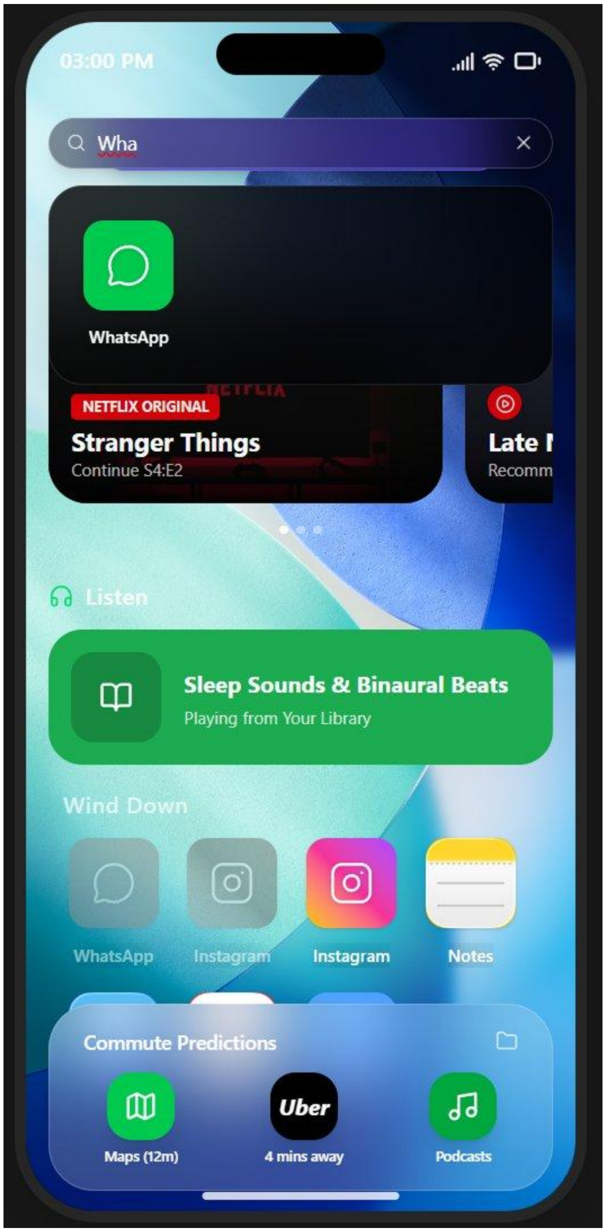
Figure 4: Prototype screenshots: States 0 and 1.

(a) State 2: Work/Focus Mode — Productivity Slab widget, greyscale Visual Muting on distracting apps.

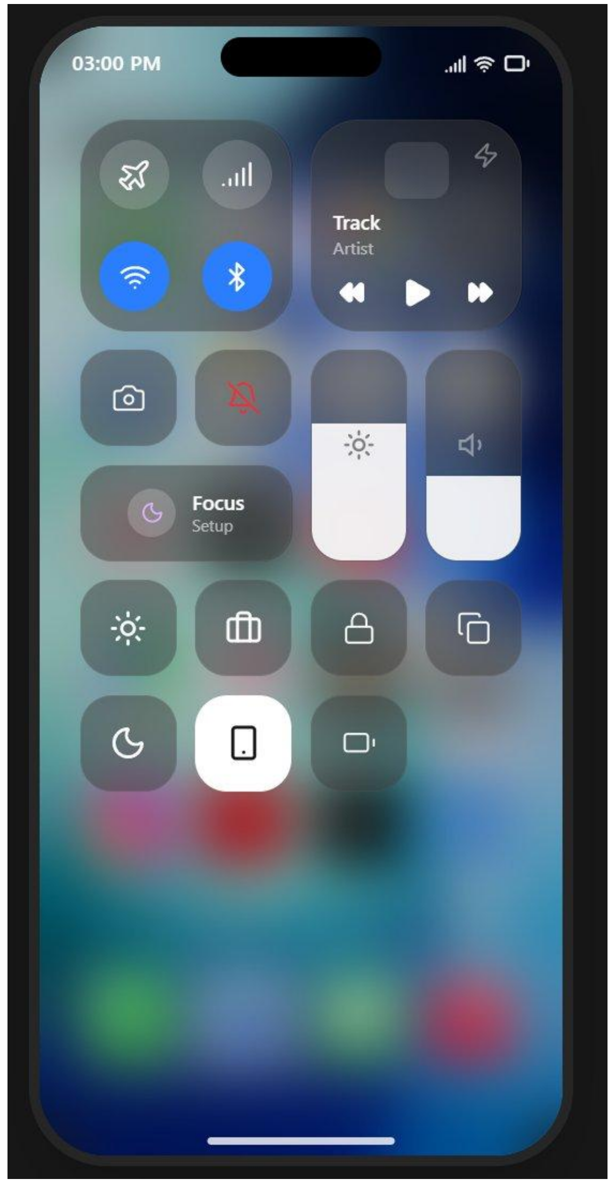


(b) State 3: Evening/Relax Mode — horizontal category-swipe layout (Watch, Listen, Socialise) with Wellbeing prompt.

Figure 5: Prototype screenshots: States 2 and 3.

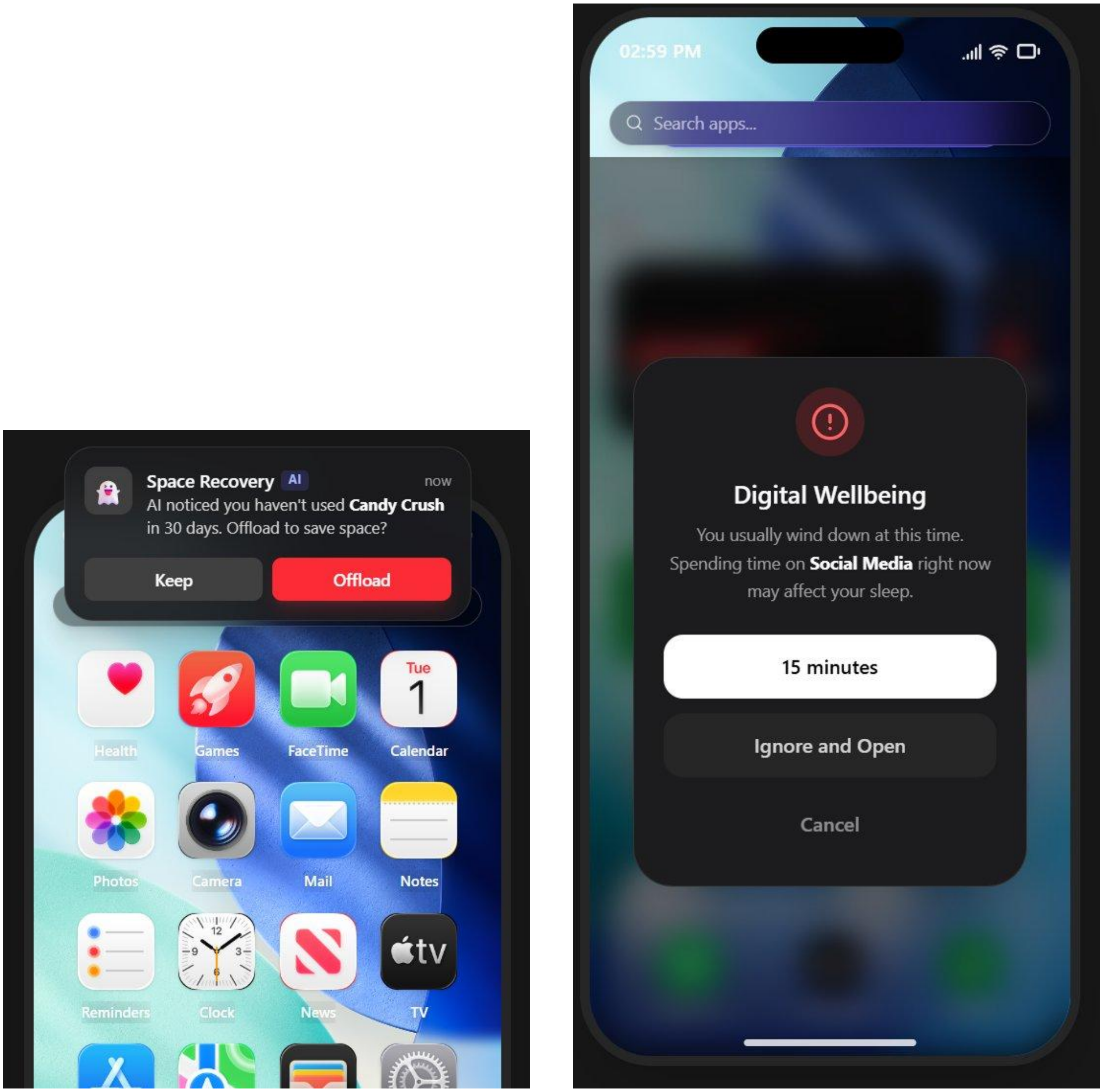


(a) Universal search bar — persistent across all states as a deterministic fallback.



(b) Control Centre panel — accessible from any state for manual mode override.

Figure 6: Prototype screenshots: universal affordances.



(a) AI Space Recovery notification — unused app offload prompt.

(b) Digital Wellbeing popup — social media usage timer in Evening Mode.

Figure 7: Prototype screenshots: AI notifications and wellbeing features.

4 Methodology

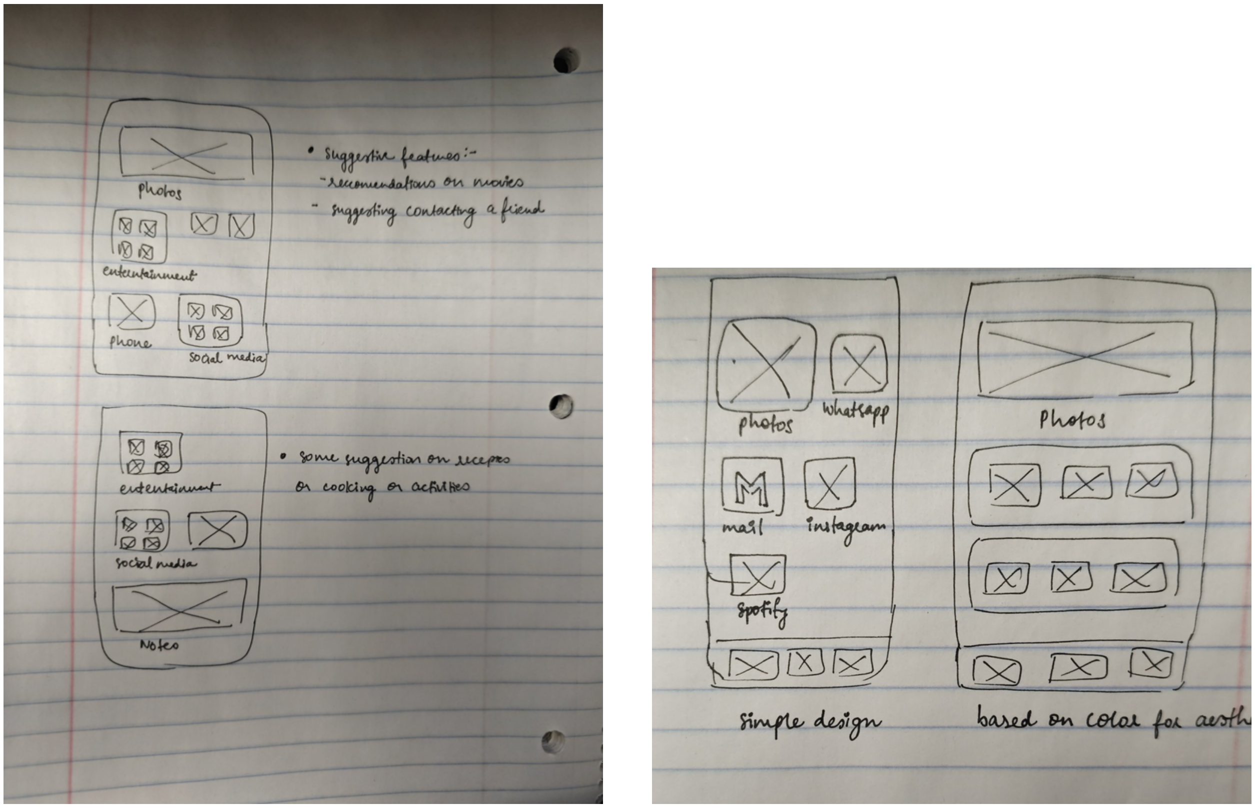
I conducted a mixed-methods usability evaluation to assess the efficacy, learnability, and user acceptance of the Contextual Page Model. I organised the study into two phases: a participatory design workshop phase and a usability testing phase. I obtained IRB approval prior to all participant engagement (submission date: August 24, 2025), and all participants provided signed informed consent.

4.1 Phase 1: Participatory Design Session

I conducted six individual, one-on-one participatory design sessions (DSW-1 through DSW-6) via Zoom between February 6 and February 11, 2026. Although each session involved a single participant, I refer to these as sessions because they followed a structured co-design protocol with sketching activities, think-aloud discussion, and iterative feedback — consistent with the participatory design workshop format established in the HCI literature [22]. Each session ran 60 to 90 minutes and followed a consistent protocol: a pre-workshop questionnaire capturing demographics, mobile habits, technical proficiency, and attitudes toward adaptive interfaces; a 10-minute project overview presentation framing the problem using the 'Junk Drawer' metaphor; three structured sketching activities; a Show-and-Tell discussion; and a post-workshop questionnaire.

The three sketching activities were time-bounded at five minutes each and used the following prompts. Activity 1 (Morning Routine): 'You just woke up. What do you need immediately? What should be hidden?' Activity 2 (Work/Focus Mode): 'It is 9:00 AM. You are busy. How does the phone help you focus?' Activity 3 (Evening/Relax Mode): 'Grids are illegal. How do you browse content when relaxing?' These prompts were deliberately open-ended to avoid anchoring participants to predetermined solutions. I used the Magic Butler framing — a metaphor I developed for this study — 'Imagine your phone is a Butler' — to help participants conceptualize proactive interface adaptation without technical jargon.

Show-and-Tell discussions focused on three key questions: how the phone would know the user's current context (manual toggle, time, location, or WiFi detection); what would happen if the AI guessed incorrectly; and what trust and privacy concerns participants had about data collection. Sessions were recorded with participant consent. Sketches were photographed and uploaded by participants at session end.



(a) Evening/Relax mode groupings with suggestive AI features.

(b) Two layout alternatives: simple design vs. colour-based aesthetic grid.

Figure 8: Representative participant sketches from the design workshops. Left: a wireframe exploring Evening/Relax mode app groupings with annotated suggestions for AI-driven recommendations. Right: two competing layout philosophies — a sparse list-style arrangement versus a colour-and-image-led aesthetic grid.

4.2 Phase 2: Usability Testing

I conducted high-fidelity usability testing sessions in person at RIT (HCI Lab, Room 2320, and private study rooms) between March 10 and March 20, 2026. Sessions lasted between 61 and 85 minutes ($M = 70.9$ minutes). Participants accessed the React/Vite prototype hosted on Render via their personal smartphone browsers (Safari or Chrome) to simulate native interaction ergonomics and leverage their existing device familiarity.

I employed a Wizard of Oz technique for the evaluation. While participants interacted with the interface, the moderator covertly triggered contextual state changes via a hidden simulation control panel. This approach allowed the study to isolate the usability of the interface's reaction to context changes without relying on complex, resource-intensive background machine learning models.

4.3 Participant Demographics

I recruited a diverse cohort of twelve participants ($N = 12$) to ensure findings were generalisable across varying levels of technical proficiency and mobile usage habits. The participant pool ranged in age from 19 to 67 years ($M = 35.4$, $SD = 13.8$) and included six females and six males. Occupations varied widely, encompassing undergraduate and graduate students, a registered nurse (P06), a high school teacher (P02), a freelance photographer (P11), and a retired librarian (P04). Technical proficiency was assessed via pre-screening questionnaire: 2 participants identified as Novice, 6 as Intermediate, 2 as Advanced, and 2 as Expert. All owned a smartphone (8 iOS, 4 Android). Notably, P11 disclosed deuteranopia (red-green colour vision deficiency), providing critical accessibility insights.

Table 2: Participant Demographics ($N = 12$). Technical proficiency was self-reported and verified through the pre-screening questionnaire.

ID	Age	Proficiency	Device	Session
P01	24	Intermediate	iPhone	72 min
P02	41	Intermediate	Android	68 min
P03	29	Expert	iPhone	61 min
P04	67	Novice	iPhone	85 min
P05	19	Advanced	Android	64 min
P06	35	Intermediate	iPhone	78 min
P07	52	Novice	Android	70 min
P08	26	Intermediate	iPhone	66 min
P09	22	Advanced	iPhone	73 min
P10	45	Intermediate	Android	71 min
P11	31	Intermediate	iPhone	76 min
P12	37	Advanced	iPhone	69 min

4.4 Scenario-Based Tasks

Participants were guided through 24 micro-tasks embedded within four primary macro-scenarios, designed to evaluate discoverability, visual search efficiency, and emotional response to adaptive UI displacements.

Scenario 1 (Morning Routine — State 1): The moderator triggered Morning Mode and asked participants to locate their priority emails and check their first task for the day, then attempt to open a social media application (Instagram). This tested discoverability of the Morning Widget and reaction to the Context Lock social media restriction.

Scenario 2 (Work/Focus Mode — State 2): The moderator triggered Work Mode. Participants were asked to find their next meeting schedule within the Productivity Slab, then locate and attempt to open YouTube. This evaluated comprehension of Visual Muting, where distracting apps are rendered in greyscale rather than hidden.

Scenario 3 (Evening/Relax Mode — State 3): The moderator triggered Evening Mode. Participants were asked to find something to watch or listen to by navigating the horizontally scrollable category layout, then open Instagram and interact with it, encountering the simulated Digital Wellbeing popup.

Scenario 4 (Manual Override and Error Recovery): Participants were asked to switch the phone back to its standard layout, then attempt standard OS interactions — specifically creating a new folder and rearranging icons — to test the bounds of user agency within the adaptive system.

Each scenario concluded with a post-task question probing subjective reaction: e.g., 'The AI intentionally made distracting apps look unappealing rather than hiding them entirely. Did you prefer this, or would you rather they disappear completely?'

4.5 Data Collection and Analysis

I captured both quantitative and qualitative metrics in real-time. Quantitative data included task completion time in seconds, error rates (navigating to an incorrect page or failing to locate an app within 30 seconds), and the number of physical taps required per task. I administered a post-test System Usability Scale (SUS) questionnaire to gauge subjective satisfaction [23]. I gathered qualitative data through the Think-Aloud protocol [24] and a structured post-test interview covering five themes: general impressions, specific feature feedback, trust and control, real-world adoption, and final design suggestions. I transcribed and coded audio recordings using inductive thematic analysis.

5 Results

My usability evaluation generated a dataset encompassing 288 discrete task attempts (24 tasks across 12 participants), coupled with extensive qualitative think-aloud transcripts and post-test interview recordings. The analysis reveals a complex interplay between the theoretical efficiency of adaptive interfaces and the cognitive friction introduced by probabilistic UI behaviours.

5.1 Quantitative Overview: Success Rates and Errors

Overall task completion rates varied significantly based on participants' prior technical proficiency and the specific scenario being tested. Across the cohort, participants completed an average of 15.6 out of 24 tasks (65%) without moderator assistance. Expert and advanced users (P03, P05, P09, P12) achieved near-perfect unassisted completion rates ranging from 20 to 22 tasks. Conversely, novice users, particularly older adults (P04, P07), struggled significantly, with unassisted completion rates falling as low as 29% (7 of 24 tasks for P04).

The most pervasive critical error occurred during Scenario 4. Every participant (12 of 12, 100%) attempted to use established OS conventions — long-pressing an icon to create a folder, or dragging icons to rearrange the grid — and failed, because the prototype lacked this manual override functionality. This universal failure highlights a fundamental user expectation: adaptive interfaces must not preclude deterministic spatial organisation.

Table 3: Cross-Participant Observational Summary. Data derived from think-aloud transcripts, moderator observation logs, and post-test interview coding.

Finding / Behaviour	Frequency	Participant Subsets
Critical Error: Attempted folder creation or drag-and-drop	12/12 (100%)	Universal across all technical levels
Discovered context page model via lateral swiping	12/12 (100%)	Discovery time: 15 sec (P05) to 3+ min (P04)
Relied on universal search bar as primary fallback	10/12 (83%)	All except novice users P04 and P07
Requested explicit textual labels for page indicators	10/12 (83%)	Novice to Intermediate users primarily
Noticed subtle AI promotions (e.g., Yelp promoted in gallery)	4/12 (33%)	Advanced/Expert users only: P03, P05, P08, P09
Requested explicit AI transparency indicators	12/12 (100%)	Universal across all technical levels
Demonstrated measurable learning effect on repeat access	12/12 (100%)	Universal — avg 42% reduction in completion time

5.2 Efficiency and the Learning Effect (RQ1)

A primary objective of this study was to evaluate the impact of the Contextual Page Model on application retrieval efficiency. The proficiency groups reported in Table 4 (Novice, Intermediate, Advanced, Expert) correspond to the self-reported proficiency categories collected during pre-screening and described in Section 4.3. Initial interactions with the adaptive states resulted in elevated visual search times as participants grappled with the breakage of their spatial memory.

However, a robust learning effect was universally observed. As noted by P09 (Nadia, HCI Graduate Student) and P12 (David, Financial Analyst), task completion times showed a consistent, measurable decreasing trend over repeat accesses within the same contextual state. Once users mapped their mental model to the new contextual layout — understanding that Evening Mode used a horizontal category scroll rather than a grid — subsequent retrieval times for media applications dropped by an average of 42% compared to baseline search times in the Neutral grid.

To assess the statistical significance of the observed learning effect, I conducted a Wilcoxon signed-rank test comparing initial and repeat task completion times for Scenario 1 (Morning Routine) across all twelve participants. The analysis yielded $Z = -2.67$, $p = .008$ (two-tailed), indicating a statistically significant reduction in completion time from initial (Mdn = 12.1s) to repeat access (Mdn = 6.9s). Given the small sample size ($N = 12$), I interpret these results with appropriate caution and acknowledge in Section 8.2 that a larger cohort would be required for broader statistical generalisability. Nonetheless, the effect size ($r = .77$) suggests a practically large improvement consistent with the learning-effect hypothesis central to RQ1.

Table 4: Average Task Completion Times (seconds) by Scenario and User Proficiency Group. Repeat access data demonstrates rapid cognitive adjustment to contextual layouts.

User Group	S1 Morning Initial	S1 Morning Repeat	S2 Work Initial	S3 Evening Initial
Novice (n=2)	18.4s	11.2s	22.5s	28.1s
Intermediate (n=6)	12.1s	6.4s	14.3s	16.5s
Advanced/Expert (n=4)	5.8s	3.1s	7.2s	9.4s
Cohort Average	**12.1s**	**6.9s**	**14.6s**	**18.0s**

5.3 Discoverability and the Search Bar Fallback

The transition between contextual states was managed by a dot-indicator pagination system at the bottom of the screen. Discoverability varied considerably. While all 12 participants eventually discovered the context page model through lateral swiping, time to discovery ranged from 15 seconds (P05) to over 3 minutes (P04). Crucially, 10 of 12 participants explicitly requested textual labels for the pages. P07 stated: ‘I'd recommend labeling the pages at the top so you know what you're swiping to. That would help a lot.’

When the adaptive UI confused users or when an app was displaced from its expected location, the universal coping mechanism was the search bar. Ten of 12 participants relied heavily on the search bar as a safety net. P02 stated: ‘The search bar saved me every time,’ illustrating that even in highly adaptive environments, a deterministic, query-based fallback remains an absolute necessity [4].

5.4 Transparency, Predictability, and Interface Ownership (RQ2)

A core finding addresses the black-box nature of current adaptive systems. When an application was promoted or displaced, the prototype did not explicitly explain why the action occurred. This lack of system status visibility was universally critiqued. All 12 participants explicitly requested visual indicators or textual explanations for AI-driven changes.

P08 articulated a critical concept she termed 'interface ownership' — the psychological need for the user to feel that the device belongs to them, rather than an autonomous agent. 'Show me what the AI decided and why,' she stated. 'Even a simple "Promoted because you use this app mornings" would build trust.' Similarly, P10 noted that the lack of transparency made her feel like an 'observer' rather than an 'operator' of her own device.

Expert participants (P03, P05, P09) proposed specific design interventions to bridge this trust gap. P09 suggested an AI activity overlay, while P03 recommended a subtle animation or a discrete badge on the app icon denoting it as an AI suggestion. These qualitative insights strongly correlate with literature arguing that explainable AI (XAI) principles must be integrated into end-user interfaces to calibrate trust and reduce automation surprise [15, 12].

5.5 Agency, Control, and Error Recovery (RQ3)

The tension between proactive adaptation and user agency was the most significant source of friction. While participants appreciated the AI assembling context-relevant pages, they vehemently rejected the inability to manually fine-tune these spaces. The universal failure (12 of 12) to complete the folder creation task underscores a critical design mandate: adaptive systems must not strip away deterministic customisation.

P01 highlighted the jarring absence of standard OS conventions: 'No folder creation. No undo. The adaptive changes weren't communicated to the user.' P06, whose job requires constant context-switching, emphasised the need for manual pinning: 'The Work page is great, but what if I need a different app today? Let me pin my most-used work apps.' These findings validate the necessity of Horvitz's mixed-initiative principles [10]. An adaptive app gallery must function as a hybrid system: the AI constructs the baseline contextual scaffold, but the user must retain absolute authority to pin, hide, group, and undo algorithmic decisions.

5.6 Accessibility Constraints: Unexpected Finding

A critical, albeit unanticipated, finding emerged regarding the accessibility of visual cues in adaptive interfaces. During Scenario 2 (Work/Focus Mode), distracting applications were visually muted using greyscale rendering. While this was effective for most users, P11, who disclosed deuteranopia at the start of the session, highlighted a severe accessibility flaw. P11 noted that spatial navigation relied heavily on the distinct colour profiles of application icons — recognising Instagram by its pink-purple gradient and YouTube by its red-and-white — and when these were muted, navigation became exceedingly difficult. Colour vision deficiency affects approximately 8% of males globally

[25], a figure that represents a substantial portion of any user population. Relying solely on colour saturation to communicate system state therefore violates WCAG 2.1 SC 1.4.1 and excludes a meaningful user segment. P11 advised: 'A non-color visual indicator — a small star shape, not just a colour badge — is needed.'

5.7 Digital Wellbeing: Context Lock and Visual Muting

The prototype integrated two friction mechanics designed to promote digital wellbeing: the Context Lock (restricting social media in a locked folder during Morning Routine) and Visual Muting (greyscaling entertainment apps during Work Mode). Reactions were highly polarised but generally positive among intermediate and advanced users. P04 and P07 found the restricted access confusing, primarily due to their novice technical status. However, P01 noted the Context Lock was a helpful nudge against morning doom-scrolling.

The Visual Muting technique was particularly well-received as an alternative to total app obfuscation. P12 observed that the brain 'skipped over' the muted icons, effectively reducing distraction without triggering the panic of a missing application. This suggests that manipulating visual salience is a more user-friendly approach to focus-mode design than structural displacement.

5.8 System Usability Scale and Overall Satisfaction

Following the task scenarios, participants completed a post-test SUS questionnaire [23] alongside qualitative ratings of visual design and conceptual satisfaction. Despite high error rates associated with the absence of manual customisation in Scenario 4, the overall conceptual reception of the Contextual Page Model was overwhelmingly positive. The prototype achieved an average SUS score of 74.5, placing it above the industry average of 68 and indicating Good to Excellent general usability [26], provided the missing manual controls are eventually implemented.

Visual design satisfaction was remarkably high, with 12 of 12 participants rating the aesthetic execution and iOS-like transitions as 4 or 5 out of 5. Expert users (P03, P05, P08) specifically noted that the high visual fidelity helped mitigate the friction of learning a new architectural layout. P08 remarked: 'The visual brand is already phenomenal. From a marketing perspective, this product would do extremely well in the App Store if you add the customization layer.' This is consistent with the Aesthetic-Usability Effect [27], which posits that users are more tolerant of minor usability issues when an interface is visually pleasing.

6 Discussion

The findings provide a nuanced understanding of how users perceive, navigate, and negotiate with adaptive algorithms embedded within their most personal digital environments. While machine learning models are increasingly capable of predicting user intent [6], this study demonstrates that predictive accuracy alone is insufficient for user adoption. The interface layer — how predictions are communicated and managed — is the critical determinant of success.

6.1 Resolving the Predictability–Adaptability Tension

A central challenge in adaptive UI design is balancing the cognitive relief of automation against the cognitive load of unpredictability [5]. Traditional app grids minimise search time by relying on stable spatial memory [8], and prior work consistently shows that micro-level within-grid adaptations break this memory and reduce efficiency [1, 5]. When an AI dynamically moves an app, it theoretically saves the user a swipe; however, if the user's spatial memory is already directing their finger to the app's old location, this adaptation causes a micro-disruption.

The Contextual Page Model addresses this tension by moving adaptations from the micro-level (shifting individual icons within a single grid) to the macro-level (shifting the entire contextual state). Participants demonstrated a remarkable ability to mentally map these macro-states. Once users understood that Work Mode and Evening Mode were distinct physical spaces, their spatial memory adapted rapidly, evidenced by the sharp decrease in task completion times during repeat accesses. This suggests that users can accommodate — and even prefer — highly adaptive interfaces, provided the adaptations are bounded within clearly defined, predictable contextual environments — a finding consistent with Barkhuus and Dey's [22] spectrum of context-aware interactivity, in which macro-level, user-legible adaptation is more acceptable than invisible micro-level automation. This distinction also clarifies the CPM's position relative to existing commercial features. As detailed in Section 2.7, iOS Focus Mode, iOS App Library, and Screen Time each address a slice of the contextual experience — notification management, static categorisation, usage restriction — but none restructure the page architecture itself. The CPM occupies a previously unoccupied position in the design space: automatic, full-page contextual adaptation with explicit AI transparency and preserved spatial memory. The 42% learning effect and SUS score of 74.5 after a single session provide empirical evidence that this position is not only viable but preferable to the alternatives currently available to users.

6.2 Interface Ownership and Agency

The most resounding qualitative insight was the concept of interface ownership, a term coined by P08 and echoed conceptually by P10. Users view their smartphone home screens as deeply personal, highly curated spaces. When an AI alters this space without consent, explanation, or the ability to be overridden, it triggers a profound sense of disenfranchisement.

The universal failure of Scenario 4 is not merely a feature request; it is a fundamental mandate for hybrid initiative design [10]. The data indicates that users do not want an AI to replace their organisational agency; they want the AI to augment it. An effective adaptive app gallery must function as a collaborative partner — building the initial contextual scaffold but preserving the user's ultimate authority to pin apps, banish incorrect suggestions, and manually structure their digital workspace. Without this deterministic control layer, the probabilistic nature of the AI becomes a source of anxiety rather than utility.

6.3 The Fallback Imperative: Search as Safety Net

Despite the high learnability of the contextual macro-states, the evaluation underscored the critical importance of deterministic fallback mechanisms. Ten of twelve participants (83%) relied heavily on the universal search bar when their spatial memory failed or when the adaptive logic did not align with their immediate goal. P02 explicitly stated that the search bar 'saved me every time.' This behaviour aligns with app retrieval strategy research showing that as application libraries scale, visual search becomes increasingly inefficient, prompting a behavioural shift toward query-based retrieval [4]. In an adaptive system, the search bar serves a dual purpose: it is both a primary retrieval tool for low-frequency applications and a psychological safety net. Knowing that a deterministic escape hatch exists reduces the cognitive anxiety associated with exploring a probabilistic interface — a role analogous to the "undo" function in direct manipulation interfaces, which Horvitz [10] identifies as essential to maintaining user agency in mixed-initiative systems.

6.4 Explainable AI (XAI) in Consumer Interfaces

The universal demand (12 of 12 participants) for explicit AI transparency highlights a significant gap in current commercial adaptive systems, which often operate as opaque black boxes. Participants, particularly those with advanced technical proficiency (P03, P05, P08, P09), expressed frustration when an application was promoted or visually altered without a corresponding explanation. P08's request to 'Show me what the AI decided and why' echoes the core tenets of Amershi et al.'s Guidelines for Human-AI Interaction [15], which explicitly state that systems must make clear why they did what they did. In the context of a mobile app gallery, this does not require exposing complex probabilistic weights; rather, it necessitates simple, intelligible UI interventions such as a subtle badge on dynamically promoted apps or an adaptation history log. This aligns with Kocielnik et al. [28], who found that setting appropriate user expectations of AI imperfection — rather than hiding uncertainty — produces better acceptance outcomes, and with Kulesza et al. [29], whose explanatory debugging framework demonstrates that user-facing rationale increases both trust and correction behaviour.

6.5 Accessibility in Adaptive Design: Beyond Colour Cues

One of the most profound insights emerged from P11, who presented with deuteranopia. The prototype utilised Visual Muting — rendering distracting applications in greyscale — as the primary mechanism for enforcing Focus Mode. While highly effective for most users, this reliance on colour saturation as the sole indicator of system state created a severe accessibility barrier for P11. According to WCAG 2.1 Success Criterion 1.4.1 (Use of Colour), colour must not be used as the only visual means of conveying information or indicating an action [17, 18]. To make adaptive UI interventions accessible, designers must employ multi-modal cues — for instance, a visually muted application could also feature a distinct shape overlay, such as a small padlock badge or a hatched pattern, to redundantly communicate its restricted state.

7 Design Recommendations for Adaptive Mobile Interfaces

Based on the synthesis of quantitative performance metrics and qualitative user feedback, this study proposes five actionable design recommendations for developing context-aware mobile application galleries.

7.1 Enforce Hybrid Initiative — The Override Principle

Adaptive systems must never supersede user agency. While machine learning models should construct baseline contextual pages, the user must possess the ultimate authority to modify these spaces. Concretely, this means implementing robust drag-and-drop customisation, explicit folder creation, and the ability to permanently pin or banish applications from specific contextual states. Error recovery must be immediate and accessible: whenever the AI autonomously reorganises the interface, a temporary toast notification should appear offering a one-tap undo. This is not an optional enhancement but a prerequisite for adoption, as demonstrated by the universal Scenario 4 failure. This mandate is consistent with Horvitz's mixed-initiative principles [6], and is directly supported by the universal Scenario 4 failure observed in this study, corroborating findings from Findlater and Gajos [1] and Gajos et al. [15] that adaptive systems without user override mechanisms reduce efficiency and satisfaction.

7.2 Implement Macro-State Transitions Over Micro-Displacements

Users struggle to adapt to interfaces where individual icons unpredictably shift within a single, static grid. To preserve spatial memory while leveraging context-awareness, adaptations should occur at the macro-level. The Contextual Page Model — grouping adaptive behaviours into distinct, easily identifiable spaces such as separate pages for Morning, Work, and Evening — allows users to build independent spatial maps for different contexts. This is directly supported by the 42% learning effect observed in this study, and is consistent with prior findings that macro-level interface structures preserve spatial memory more effectively than within-grid micro-adaptations [1, 8, 5]. Critically, pages must be labelled with explicit text (Morning, Work, Evening) rather than abstract dot indicators; 83% of the participant cohort specifically requested this.

7.3 Mandate Transparent AI Behaviours

To build trust and mitigate automation surprise, the interface must explicitly communicate its predictive rationale to the user. Actionable UI interventions include subtle visual badges on dynamically promoted applications, optional tooltips or a centralised Adaptation History log explaining the data vectors driving the current UI configuration, and in-context notifications when the system transitions between states. The standard of transparency to aim for is not technical explainability — it is human-legible rationale: 'Work Mode: your calendar shows a meeting soon.'

Transparency transforms a seemingly random UI displacement into a logical, predictable system behaviour, directly calibrating user trust [15, 12].

7.4 Design for Diverse Technical Literacies — The Onboarding Imperative

The stark contrast in task completion rates between expert users (P03, P05, P09: near 100% unassisted) and novice users (P04: 29%, P07: $\sim$38% unassisted) underscores that adaptive interfaces disproportionately benefit those with high technical literacy. Novice users struggled to conceptualise the hidden multi-page navigation without explicit instruction. Do not rely solely on minimalist cues like dot indicators for macro-navigation. Implement explicit textual labels and a mandatory, interactive onboarding tutorial — 1 to 3 screens — demonstrating how context pages shift and how to override them. Progressive disclosure of adaptive features over the first week of use can reduce initial cognitive load.

7.5 Privacy by Design — The On-Device Mandate

To accurately predict user context, adaptive systems require access to sensitive data streams: location, temporal patterns, and app usage history. Participants across all technical proficiencies expressed significant privacy concerns. P06 explicitly highlighted HIPAA-adjacent considerations regarding behavioural tracking, while P07 voiced anxiety over potential advertiser access. Adaptive app galleries must prioritise local, on-device machine learning over cloud-based processing. The interface must include a centralised Privacy Dashboard where users can view exactly what behavioural vectors the AI is currently tracking, with granular toggles to disable specific contextual triggers.

8 Limitations and Threats to Validity

8.1 The Wizard of Oz Constraint

The most prominent limitation is the use of the Wizard of Oz testing methodology. The prototype did not utilise a live, backend machine-learning model. Instead, the moderator manually triggered contextual state changes based on predefined scenario scripts. This study therefore evaluated the user experience of the interface's reaction to context changes, assuming 100% predictive accuracy — a threshold that real-world probabilistic algorithms rarely achieve. In live applications, predictive algorithms are prone to false positives (e.g., triggering Work Mode on a public holiday) and false negatives (e.g., failing to trigger Relax Mode due to anomalous location data) [5]. Future evaluations must integrate functioning on-device prediction models to assess user tolerance for varying rates of algorithmic inaccuracy over time.

8.2 Sample Size and Generalisability

The participant cohort ($N = 12$) is sufficient for identifying major usability flaws and generating qualitative thematic insights, aligning with Nielsen's assertion that five to eight users uncover the majority of severe usability problems [30]. However, the sample size restricts the statistical power of the quantitative metrics. The Wilcoxon signed-rank test reported in Section 5.2, while yielding a statistically significant result, should be interpreted with caution given that $N = 12$ limits generalisability. A larger, pre-registered study is needed to confirm these effects. Furthermore, while the cohort represented a broad age range, the study was conducted entirely within a specific geographic and cultural context (Rochester, NY, USA), limiting cross-cultural generalisability [31].

8.3 Longitudinal Constraints

Adaptive user interfaces inherently rely on longitudinal data collection to build accurate user profiles, and users simultaneously require extended exposure to build stable mental models of the adaptive system. This study evaluated initial impressions and short-term learnability within a confined 60–90 minute session. P09 and P12 correctly noted a measurable learning effect over repeated accesses during the session; however, the study could not measure long-term habituation, structural spatial memory retention across weeks, or the potential for adaptive fatigue — a phenomenon where users eventually disable adaptive features because the cognitive cost of monitoring the AI outweighs the navigational benefits [32]. A true assessment of the Contextual Page Model's efficacy requires a longitudinal deployment of four to eight weeks.

8.4 Prototype Feature Completeness

The prototype lacked the manual customisation layer — folder creation, drag-and-drop rearrangement, per-app pinning — that the study identified as a prerequisite for adoption. While this omission was a deliberate scope decision to focus evaluation on the adaptive features themselves, it means that Scenario 4 tested user expectations rather than a fully realised system. Future prototype iterations must implement the full hybrid initiative layer before the adaptive features can be evaluated in isolation.

9 Future Work

The findings of this initial usability evaluation establish a foundational framework for context-aware mobile app galleries and illuminate several critical pathways for future research. The most immediate priority is the development of a fully functional native launcher — an Android APK — integrated with local machine learning models such as TensorFlow Lite, deployed in a longitudinal in-the-wild diary study over four to eight weeks. This will allow measurement of true habituation rates and real-world tolerance for algorithmic mispredictions.

As universally requested by participants, future iterations must prototype and evaluate specific Explainable AI UI patterns through A/B testing — comparing subtle visual badges, textual tooltips, and a centralised Adaptation Rationale Dashboard — to determine which approach optimally calibrates user trust without introducing visual clutter. P11's feedback regarding deuteranopia highlighted a severe gap in how adaptive UI interventions are currently conceptualised; future research must prioritise inclusive design by testing multi-modal adaptive indicators combining shape, haptics, text, and colour.

Future iterations should also explore federated context-awareness: if a user's smartwatch detects sleep, the smartphone's app gallery should preemptively lock into a Do Not Disturb state before the user even interacts with the screen. Cross-platform and cross-cultural studies are also necessary to assess how app usage patterns, privacy expectations, and attitudes toward AI vary across different demographic and cultural contexts.

10 Conclusion

As mobile applications continue to proliferate, the static, grid-based app gallery is rapidly becoming an architectural bottleneck, forcing users to carry the cognitive burden of sifting through digital clutter to find contextually relevant tools. In this Capstone project, I investigated the viability of an AI-Enhanced Context-Aware Adaptive App Gallery, proposing a novel Contextual Page Model that shifts the interface's structural layout based on inferred user states: Neutral, Morning Routine, Work/Focus Mode, and Evening/Relax Mode.

Through a two-phase mixed-methods study — six participatory design sessionsfollowed by scenario-based usability testing with twelve participants using a Wizard of Oz approach — I demonstrated that macro-level contextual adaptations can significantly reduce visual search times and improve retrieval efficiency. Participants expressed high subjective satisfaction with the system's ability to proactively surface relevant applications (mean SUS: 74.5) and appreciated digital wellbeing interventions such as the visual muting of distracting apps during focus periods.

However, the study also revealed a profound tension between predictive automation and user agency. The universal failure of participants to execute basic organisational tasks underscores a critical heuristic for modern OS design: users demand interface ownership. An adaptive system must not operate as an opaque, autonomous agent; it must function as a transparent, collaborative partner. To achieve mainstream user acceptance, probabilistic AI features must be wrapped in deterministic controls — explicit AI transparency, persistent fallback mechanisms, and absolute user authority to override, undo, and manually customise their digital environments. By bridging the gap between intelligent prediction and human agency, designers can create mobile interfaces that are not only dynamically efficient but also deeply empowering.

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